

IP Addressing, Subnetting, ARP, and DHCP

Hosts on a local area network need both a hardware MAC address and a logical IP address. IPv4 addresses are paired with a subnet mask or CIDR prefix. The prefix tells a host whether the destination is on-link. If it is not, the packet is sent to the default gateway, usually a router or layer-3 switch on the same subnet.

Address Resolution Protocol maps an IPv4 address to a MAC address inside the LAN broadcast domain. A host sends an ARP request, the owner replies, and the mapping is cached. Stale or poisoned ARP entries cause intermittent connectivity and are a common attack technique on flat networks.

Dynamic Host Configuration Protocol assigns IP address, mask, gateway, and DNS servers automatically. A DHCP lease has a lifetime; clients renew it. Servers and printers usually receive reserved addresses so that firewall rules stay stable. Duplicate addresses produce a conflict and kick one of the hosts off the network.

Careful subnetting keeps broadcast traffic local and prepares the LAN for VLAN segmentation. A laboratory subnet, a staff subnet, and a guest subnet should not share one giant prefix. Documentation of DHCP scopes and static reservations is part of daily LAN operations.